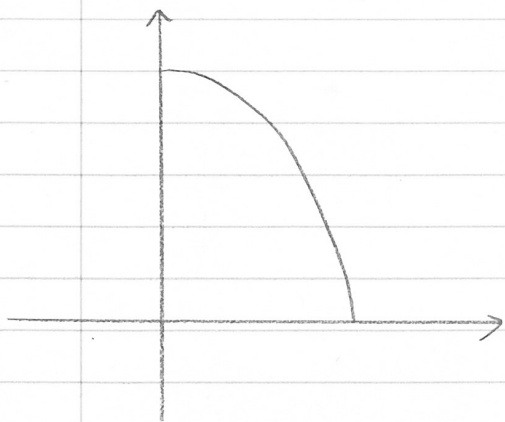
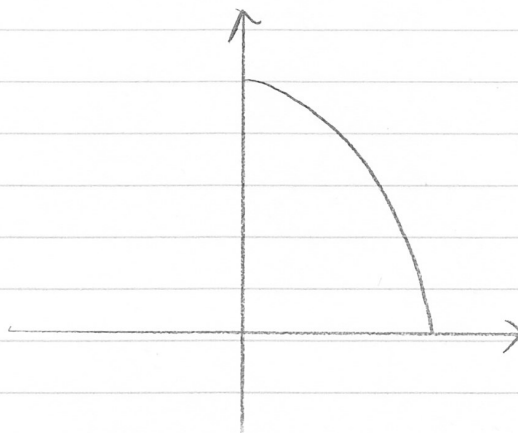


Volumes By Shells



Horizontal Rectangles



Vertical Rectangles

Theorem A shell with inner radius r_1 , out radius r_2 , and height h is said to have average radius $r = \frac{r_2 + r_1}{2}$

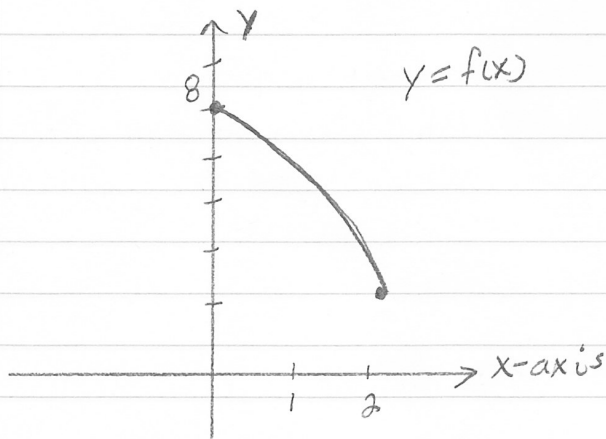
and thickness $\Delta x = r_2 - r_1$. Volume $V = 2\pi r h \Delta x$

Def. A solid of revolution obtained by rotating a region on $[a, b]$ around the y -axis has

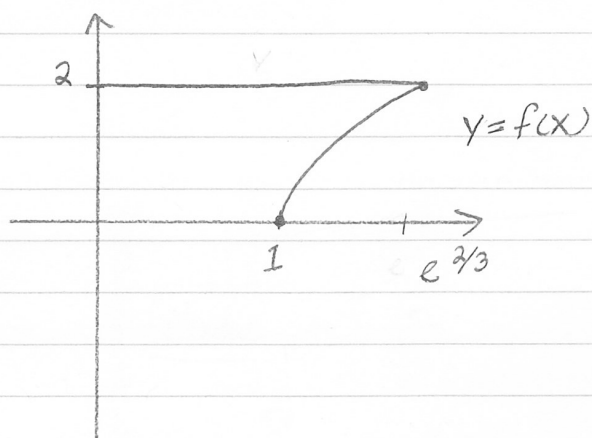
$$\begin{array}{l} \text{Volume of} \\ \text{Solid with} \\ \text{nested shells} \end{array} = \lim_{n \rightarrow \infty} \sum_{k=1}^n 2\pi r_k h_k^* \Delta x = 2\pi \int_a^b r(x) h(x) dx$$

with $r(x)$ the average radii, $h(x)$ the height.

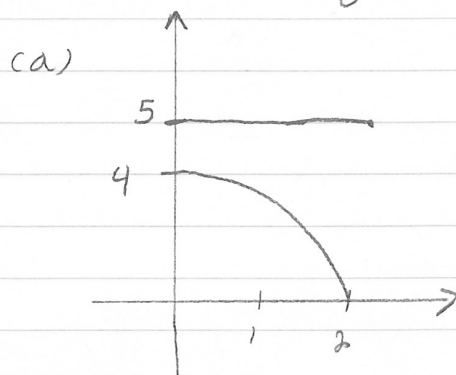
Example 1 Find the volume of the solid obtained by rotating the region between the graph of $f(x) = -x^2 - x + 8$ and the x -axis on $[0, 2]$ around the y -axis



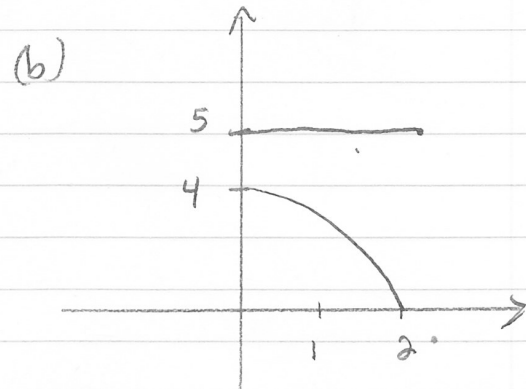
Example 2 Set up the integral to find the volume of region bounded by the graph $f(x) = 3\ln(x)$, the line $y=2$, and the x - and y -axes around the x -axis



Example 3 shell method. Set up the integral for the region between $f(x) = 4 - x^2$ and $y = 5$.



around y -axis



around $x = 2$

R is the region bounded above by the graph of $f(x) = 2x^{\frac{2}{5}}$ and below by the x -axis over the interval $[0, 6]$. Find the volume of the solid of revolution formed by revolving R around the y -axis.

Submit an exact answer in terms of π .

R is the region bounded above by the graph of $f(x) = 2 \sin(2x^2) + 2$ and below by the x -axis over the interval $[0, \sqrt{3\pi}]$. Find the volume of the solid of revolution formed by revolving R around the y -axis.

Submit an exact answer in terms of π .

Define R as the region bounded by the graphs of $f(x) = 4\sqrt{x}$ and $g(x) = x^2$ in the interval $[0, 2]$. Which of the following represents the volume of the solid of revolution formed by rotating R about the line $x = 3$?

- ☐ $\int_0^2 2\pi(3 - x) (x^2 - 4\sqrt{x}) \, dx$
- ☐ $\int_0^2 2\pi(3 + x) (4\sqrt{x} - x^2) \, dx$
- ☐ $\int_2^0 2\pi(3 - x) (4\sqrt{x} - x^2) \, dx$
- ☒ $\int_0^2 2\pi(3 - x) (4\sqrt{x} - x^2) \, dx$
- ☐ $\int_0^2 2\pi(3 + x) (x^2 - 4\sqrt{x}) \, dx$

Q is the region bounded by the graph of $v(y) = 4y$, $x = 6$, and $y = 0$. Find the volume of the solid of revolution formed by revolving Q around the x -axis.

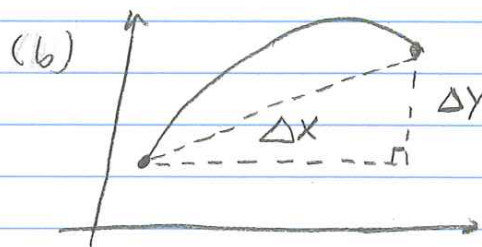
Submit an exact answer in terms of π .

Arc Length And Surface Area

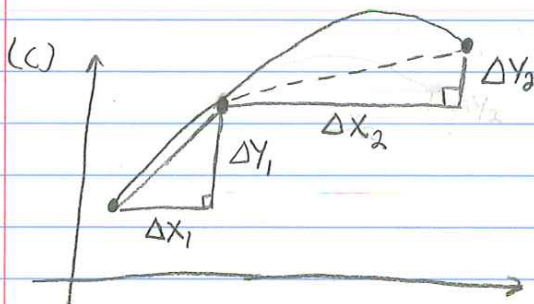
Example 1



Approx curve length l



Approx. curve length using diagonal



Approx curve length using both diagonals

2.4

Let $g(y) = 4y - 2$. Calculate the arc length of the graph of $g(y)$ over the interval $[2, 7]$.

Enter an exact answer.

Arc Length = units

2.4

Let $g(y) = \frac{y^3}{6} - 1 + \frac{1}{2y}$. Calculate the arc length of the graph of $g(y)$ over the interval $[1, 3]$.

Enter an exact answer.

2.4

Let $f(x) = 2 - \frac{x}{3}$. Calculate the arc length of the graph of $x = g(y)$ over the interval $x = 2$ to $x = 5$.

Enter an exact answer.

Theorem For $f(x)$ differentiable on $[a, b]$, the arc length of $f(x)$ from $x=a$ to $x=b$ is

$$\text{Arc length of } f \text{ from } x=a \text{ to } x=b = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x}\right)^2} \Delta x,$$

$$\text{w/ } \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x, \quad \Delta y_k = f(x_k) - f(x_{k-1}) \text{ all } k.$$

Example 2 Approximate arc length using $n=4$ line segments, $f(x) = x^3$, on $[-2, 2]$.

Example 3 Set up integral for exact arc length computation

Surface Area

Theorem. For $f(x)$ positive and differentiable,

$$\begin{array}{l} \text{Surface area of } f(x) \\ \text{from } x=a \text{ to } x=b \\ \text{revolved around } x\text{-axis} \end{array} = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx$$

Example 4 Determine the exact area for the surface of revolution obtained by $f(x) = \frac{1}{x}$ on $[1, 10]$ around the x -axis.

2.4

Let $g(y) = 4y - 2$. Calculate the arc length of the graph of $g(y)$ over the interval $[2, 7]$.

Enter an exact answer.

Arc Length = units

2.4

Let $g(y) = \frac{y^3}{6} - 1 + \frac{1}{2y}$. Calculate the arc length of the graph of $g(y)$ over the interval $[1, 3]$.

Enter an exact answer.

2.4

Let $f(x) = 2 - \frac{x}{3}$. Calculate the arc length of the graph of $x = g(y)$ over the interval $x = 2$ to $x = 5$.

Enter an exact answer.